Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

If {.} denotes the fractional part of x, the range of the function $\mathrm{f(x)} = \sqrt{\{\mathrm{x}\}^2 - 2\{\mathrm{x}\}}$ is

(A) ϕ

(B) $[0, 1/2]$

(C) $\{0, 1/2\}$

(D) $\{0\}$

Q.2

If $\mathrm{f(x)}$ is a polynomial function that $\mathrm{f(x)} \cdot \mathrm{f(-x)} = \mathrm{f(2x)}$, then-

(A) No such function exists

(B) $\mathrm{f(x)}$ is linear

(C) Number of such functions are exactly one

(D) Number of such functions are exactly two

Q.3

The number of points on the real line where the function $f(x) = \log_{|x^2-1|} |x - 3|$ is not defined is

(A) 4

(B) 3

(C) 6

(D) infinite

Q.4

The number $\mathrm{N} = \sqrt{10^{2+\left(\frac{1}{2}\right)\log 16}}$ (where base of logarithm is 10). Choose the correct option

(1) N is coprime with 9

(2) N is a simple surd

(3) N is an odd composite

(4) N forms the sides of a right triangle with 99 and 101

(A) 1 is correct

(B) 2 & 3 are correct

(C) 1 & 4 are correct

(D) 1,2 and 3 are correct

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Q.5

If the largest positive value of the function defined as
$\mathrm{f(x)} = \sqrt{8x - x^2} - \sqrt{14x - x^2 - 48}$, is $m\sqrt{n}$
where $m, n \in N, (m \neq 1)$ then the value of $(\mathrm{m} + \mathrm{n})$ is

(A) 5

(B) 6

(C) 7

(D) 9

Q.6

Consider a real valued function $\mathrm{f(x)}$ such that $\frac{1-\mathrm{e}^{\mathrm{f(x)}}}{1+\mathrm{e}^{\mathrm{f(x)}}} = \mathrm{x}$. The value of a and b for which $\mathrm{f(a)} + \mathrm{f(b)} = \mathrm{f}\left(\frac{\mathrm{a}+\mathrm{b}}{1+\mathrm{ab}}\right)$ is satisfied are -

(A) $\mathrm{a} \in (-\infty, 1), \mathrm{b} \in \mathrm{R}$

(B) $\mathrm{a} \in (-\infty, 1), \mathrm{b} \in (-1, \infty)$

(C) $\mathrm{a} \in (-1, 1), \mathrm{b} \in [-1, 1)$

(D) $\mathrm{a} \in (-1, 1), \mathrm{b} \in (-1, 1)$

Q.7

Which of the following function(s) have non-empty domain set?

(A) $f(x) = \log_{x-1}\left(2 - [x] - [x]^2\right)$

(B) $g(x) = \cos^{-1}(2 - \{x\})$

(C) $\mathrm{h(x)} = \ln \ln(\cos \mathrm{x})$

(D) None of these
(where [.] and \{.\} denotes greatest integer function and fractional part function respectively)

Q.8

Period of the function $\mathrm{f(x)} = \frac{1}{3}(\sin 3\mathrm{x} + |\sin 3\mathrm{x}| + [\sin 3\mathrm{x}])$, where [.] denotes greatest integer function

(A) $\frac{\pi}{3}$

(B) $\frac{2\pi}{3}$

(C) $\frac{4\pi}{3}$

(D) π

Q.9

Let $f(x) = g(x) + h(x)$ where $g(x) = \frac{1}{\pi}\left(\sin^{-1} x + \tan^{-1} x\right)$ and $h(x) = \frac{x+1}{x^2+2x+5}$ then -

(A) Domain of $\mathrm{f(x)}$ is $[-1, 1]$

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(B) Maximum value of $g(x)$ is $3/4$

(C) Maximum value of $h(x)$ is $1/4$

(D) Range of $\mathrm{f(x)}$ is $\left[-\frac{3}{4}, 1\right]$

Q.10

If function $\mathrm{f(x)} = \log_2\left(\frac{\sin \mathrm{x} - \cos \mathrm{x} + 3\sqrt{2}}{\sqrt{2}}\right)$ then -

(A) Domain of $f(x)$ is R

(B) Domain of $\mathrm{f(x)}$ is $\mathrm{R} - \{\mathrm{n}\pi\}, \mathrm{n} \in \mathrm{I}$

(C) Range of $\mathrm{f(x)}$ is $[1, 2]$

(D) Range of $\mathrm{f(x)}$ is $\mathrm{R} - [1, 2]$

Q.11

Which of the following statements are incorrect?

(A) If $f(x)$ and $g(x)$ are one-one then $f(x) + g(x)$ is also one-one

(B) If $f(x)$ and $g(x)$ are one-one then $f(x)g(x)$ is also one-one.

(C) If $f(x)$ is odd then it is necessarily one-one.

(D) None of these

Q.12

Passage

$\mathrm{k(x)}$ is a function such that $\mathrm{k(f(x))} = \mathrm{a} + \mathrm{b} + \mathrm{c} + \mathrm{d}$ where $a = \begin{cases} 0, & \text{if } f(x) \text{ is even} \\ -1, & \text{if } f(x) \text{ is odd} \\ 2, & \text{if } f(x) \text{ is neither even nor odd} \end{cases}$ $b = \begin{cases} 3, \text{ if } f(x) \text{ is periodic} \\ 4, \text{ if } f(x) \text{ is aperiodic} \end{cases}$ $c = \begin{cases} 5, \text{ if } f(x) \text{ is one-one} \\ 6, \text{ if } f(x) \text{ is many-one} \end{cases}$ $d = \begin{cases} 7, \text{ if } f(x) \text{ is onto} \\ 8, \text{ if } f(x) \text{ is into} \end{cases}$ $A = \{x^2, e^x, \sin x, |x|\}$ all the functions in set A are defined from R to R $B = \{18, 19, 16, 17\}$

$h : R \to R, h(x) = \left(\frac{e^{2x} + e^x + 1}{e^{2x} - e^x + 1}\right)$ and $\phi : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to R, \phi(x) = \tan x$

Question:

$\mathrm{k}(\phi(\mathrm{x}))$ is equal to -

(A) 14

(B) 16

(C) 17

(D) 18

Q.13

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Passage

$\mathrm{k(x)}$ is a function such that $\mathrm{k(f(x))=a+b+c+d}$ where $a=\begin{cases}0, & \text{if } f(x) \text{ is even}\\ -1, & \text{if } f(x) \text{ is odd}\\ 2, & \text{if } f(x) \text{ is neither even nor odd}\end{cases}$ $b=\begin{cases}3, \text{if } f(x) \text{ is periodic}\\ 4, \text{ if } f(x) \text{ is aperiodic}\end{cases}$ $c=\begin{cases}5, \text{ if } f(x) \text{ is one-one}\\ 6, \text{ if } f(x) \text{ is many-one}\end{cases}$ $d=\begin{cases}7, \text{ if } f(x) \text{ is onto}\\ 8, \text{ if } f(x) \text{ is into}\end{cases}$ $A=\left\{x^2, e^x, \sin x, |x|\right\}$ all the functions in set A are defined from R to R $B=\{18,19,16,17\}$

$h: R \rightarrow R, h(x)=\left(\frac{e^{2x}+e^x+1}{e^{2x}-e^x+1}\right)$ and $\phi:\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow R, \phi(x)=\tan x$

Question:

$\mathrm{k(h(x))}$ is equal to -

(A) 15

(B) 16

(C) 17

(D) 18

Q.14

Passage

$\mathrm{k(x)}$ is a function such that $\mathrm{k(f(x))=a+b+c+d}$ where $a=\begin{cases}0, & \text{if } f(x) \text{ is even}\\ -1, & \text{if } f(x) \text{ is odd}\\ 2, & \text{if } f(x) \text{ is neither even nor odd}\end{cases}$ $b=\begin{cases}3, \text{if } f(x) \text{ is periodic}\\ 4, \text{ if } f(x) \text{ is aperiodic}\end{cases}$ $c=\begin{cases}5, \text{ if } f(x) \text{ is one-one}\\ 6, \text{ if } f(x) \text{ is many-one}\end{cases}$ $d=\begin{cases}7, \text{ if } f(x) \text{ is onto}\\ 8, \text{ if } f(x) \text{ is into}\end{cases}$ $A=\left\{x^2, e^x, \sin x, |x|\right\}$ all the functions in set A are defined from R to R $B=\{18,19,16,17\}$

$h: R \rightarrow R, h(x)=\left(\frac{e^{2x}+e^x+1}{e^{2x}-e^x+1}\right)$ and $\phi:\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow R, \phi(x)=\tan x$

Question:

If $k(x)$ is a function such that $k: A \rightarrow B, y=k(x)$, where $x \in A, y \in B$ then $k(x)$ is -

(A) one-one onto

(B) one-one into

(C) many-one into

(D) many-one onto

Q.15

Passage

Let $\mathrm{f(x)=f_1(x)-2f_2(x)}$, where $\mathrm{f_1(x)=\min\left\{x^2,|x|\right\}}$, for $\mathrm{-1 \leq x \leq 1}$

$=\max\left\{x^2,|x|\right\}$, for $|x|>1$

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$$f_2(x) = \max\left\{x^2, |x|\right\}, \text{ for } -1 \le x \le 1$$
$$= \min\left\{x^2, |x|\right\}, \text{ for } |x| > 1$$
$$\text{and } g(x) = \begin{cases} \min\{f(t) : -3 \le t \le x, -3 \le x < 0\} \\ \max\{f(t) : 0 \le t \le x, \quad 0 \le x \le 3\} \end{cases}$$

Question:

For $-3 \le x \le -1$, range of $g(x)$ is-

(A) $[-1, 3]$

(B) $[-1, -15]$

(C) $[-1, 9]$

(D) None of these

Q.16

Passage

Let $f(x) = f_1(x) - 2f_2(x)$, where $f_1(x) = \min\left\{x^2, |x|\right\}$, for $-1 \le x \le 1$
$= \max\left\{x^2, |x|\right\}$, for $|x| > 1$
$$f_2(x) = \max\left\{x^2, |x|\right\}, \text{ for } -1 \le x \le 1$$
$$= \min\left\{x^2, |x|\right\}, \text{ for } |x| > 1$$
$$\text{and } g(x) = \begin{cases} \min\{f(t) : -3 \le t \le x, \quad -3 \le x < 0\} \\ \max\{f(t) : 0 \le t \le x, \quad 0 \le x \le 3\} \end{cases}$$

Question:

Number of critical points of $f(x)$ is -

(A) 1

(B) 2

(C) 3

(D) None of these

Q.17

Passage

Let $f(x) = f_1(x) - 2f_2(x)$, where $f_1(x) = \min\left\{x^2, |x|\right\}$, for $-1 \le x \le 1$
$= \max\left\{x^2, |x|\right\}$, for $|x| > 1$
$$f_2(x) = \max\left\{x^2, |x|\right\}, \text{ for } -1 \le x \le 1$$
$$= \min\left\{x^2, |x|\right\}, \text{ for } |x| > 1$$
$$\text{and } g(x) = \begin{cases} \min\{f(t) : -3 \le t \le x, \quad -3 \le x < 0\} \\ \max\{f(t) : 0 \le t \le x, \quad 0 \le x \le 3\} \end{cases}$$

Question:

For $x \in (-1, 0)$, $f(x) - g(x)$ is -

(A) $x^2 - 2x + 1$

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(B) $x^2 + 2x - 1$

(C) $x^2 + 2x + 1$

(D) $x^2 - 2x - 1$

Q.18

Match the column -

Column I	**Column II**
(A) Function $f:\left[0,\frac{\pi}{3}\right]\to[0,1]$ defined by $f(x)=\sqrt{\sin x}$ is	(p) one-one function
(B) Function $f:(1,\infty)\to(1,\infty)$ defined by $f(x)=\frac{x+3}{x-1}$ is	(q) many-one function
(C) Function $f:\left[-\frac{\pi}{2},\frac{4\pi}{3}\right]\to[-1,1]$ defined f (x) = sin x is	(r) into function
(D) Function $f:(2,\infty)\to[8,\infty)$ defined $f(x)=\frac{x^2}{x-2}$ is	(s) onto function
	(t) f ' (x) < 0

(A) A-r,t ; B-p,s,r ; C-q,s,r ; D-q,s,p

(B) A-p,r ; B-p,s,t ; C-q,s ; D-q,s

(C) A-r,t ; B-p,s,q ; C-p,s,t ; D-q,s,p

(D) A-p,r ; B-p,s,q ; C-p,s,r ; D-q,s,t

Q.19

If domain of $f(x)=\frac{\sin^{-1}(\sin x)}{\sqrt{-\log_{\left(\frac{x+4}{2}\right)}\log_2\left(\frac{2x-1}{3+x}\right)}}$ is $(a,b)\cup(c,\infty)$, then find the value of $a+b+3c$.

Q.20

If $f:(0,\infty)\to(0,\infty)$ satisfy $f(x\ f(y))=x^2y^a(a\in R)$, then find the value of a.

Answers & Solutions

Q.1 Answer:

$\{0\}$

Solution:

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$\{x\}^2 - 2\{x\} \geq 0$

$\Rightarrow \quad \{x\}(\{x\} - 2) \geq 0$

$\Rightarrow \quad \{x\} \leq 0$ **or** $\{x\} \geq 2$

Second case is not possible.

Hence $\{x\} = 0$

As $\{x\} \leq [0, 1)$**, hence range of** $f(x)$ **contains only one element 0.**

Q.2 Answer:

Number of such functions are exactly two

Solution:

Let degree of $\mathrm{f(x)}$ **is** n

Equating the degree of LHS and RHS, we get $n + n = n \Rightarrow n = 0$

$\Rightarrow \mathrm{f(x)} = \mathrm{c}$

$\Rightarrow \mathrm{c}^2 = \mathrm{c}$

$\Rightarrow \quad c = 0, 1 \Rightarrow \mathrm{f(x)} = 0, 1$

So, number of such functions are exactly two.

Q.3 Answer:

6

Solution:

$\log_{|x^2-1|} |x - 3|$ **is defined when** $x - 3 \neq 0$ **and** $\left|x^2 - 1\right| > 0$ **and** $\left|x^2 - 1\right| \neq 1.$

$\Rightarrow x \neq 3, x^2 - 1 \neq 0$ **i.e.,** $x \neq 1$ **or** -1

$\left|x^2 - 1\right| \neq 1 \Rightarrow x \neq -\sqrt{2}$ **or** $\sqrt{2}$ **and** $x \neq 0$

$\therefore$ **the points at which the function is not defined are** $\mathrm{x} = 0, 1, -1, \sqrt{2}, -\sqrt{2}, 3.$

Q.4 Answer:

1 & 4 are correct

Solution:

$\mathrm{N} = \sqrt{(10)^2 (10)^{\log_{10} 4}} = \sqrt{400} = 20$ **which form the sides of a right triangle with 99 and 101.**

$\left(99^2 + 20^2 = 101^2\right)$

Clearly N **is coprime with 9 .**

Q.5 Answer:

5

Solution:

We have $f(x) = \sqrt{16 - (x - 4)^2} - \sqrt{1 - (x - 7)^2}$

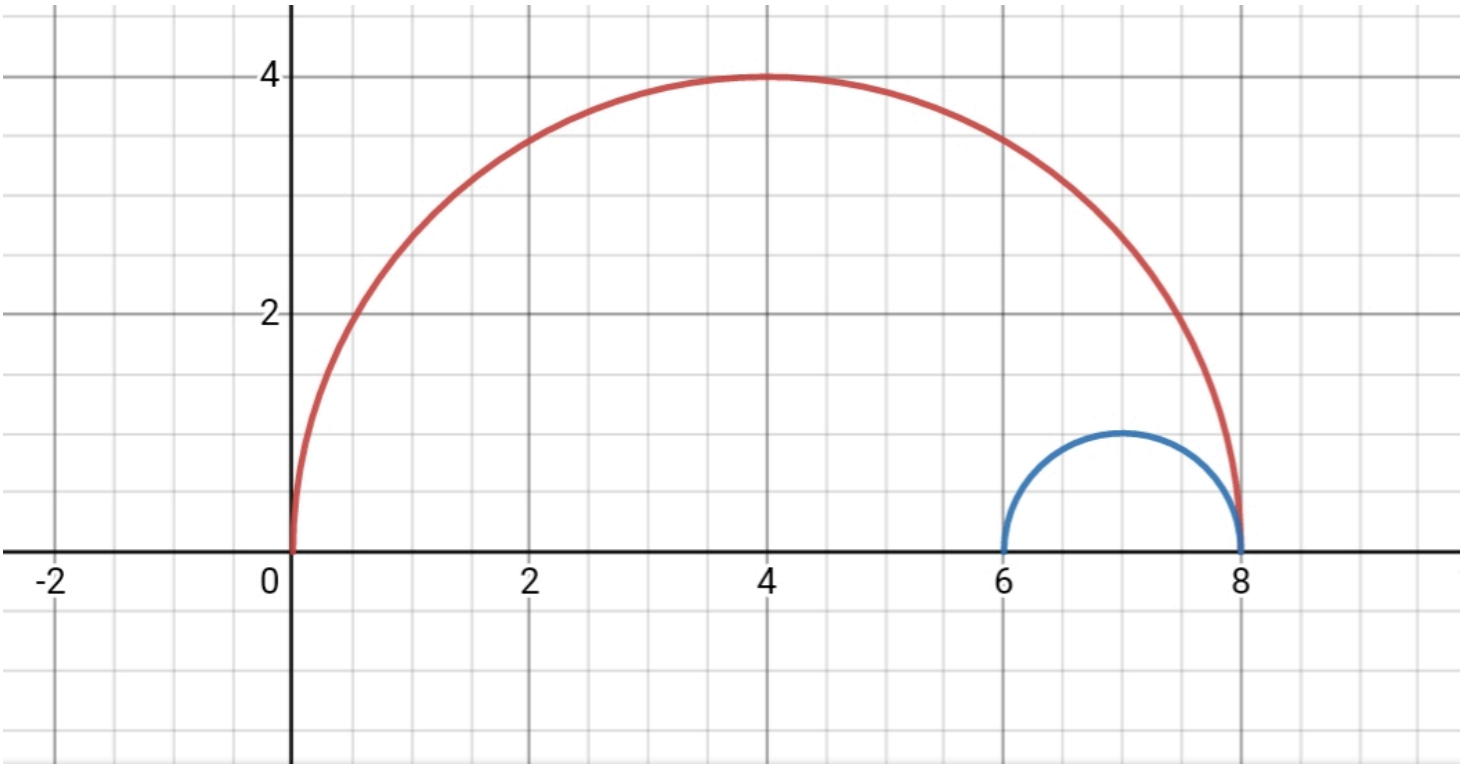
Now consider $y = \sqrt{16 - (x - 4)^2}$

$\Rightarrow (x - 4)^3 + y^2 = 16, y > 0$ **is a semi circle with centre** $(4, 0)$ **and radius** $= 4.$

Add $y = \sqrt{1 - (x - 7)^2}$

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is a semi circle with centre $(7,0)$ and radius $=1$



Now on combining the 2 figures, we have

Now $\mathrm{f(x)}]_{\max}$ **= maximum vertical distance between the 2 curves which occurs when** $\mathrm{x}=6.$

$\therefore f(x)]_{\max}=\sqrt{16-(6-4)^2}-0=\sqrt{12}$

Q.6 Answer:

$\mathrm{a}\in(-1,1),\mathrm{b}\in(-1,1)$

Solution:

Componendo-dividendo $\frac{-2\mathrm{e}^{\mathrm{f(x)}}}{2}=\frac{\mathrm{x}-1}{\mathrm{x}+1}$

$\Rightarrow \mathrm{e}^{\mathrm{f(x)}}=\left[\frac{\mathrm{x}-1}{1-\mathrm{x}}\right]$

$f(x)=\ln\frac{x+1}{1-x}$

$f(a)+f(b)=\ln\left[\frac{a+1}{1-a}\times\frac{b+1}{1-b}\right]$

$f\left(\frac{a+b}{1+ab}\right)=\ln\left[\frac{\frac{a+b}{1+ab}+1}{1-\frac{a+b}{1+ab}}\right]\ldots\ldots\ldots(2)$

Now, $\frac{(a+1)(b+1)}{(1-a)(1-b)}=\frac{a+b+1+ab}{1+ab-a-b}$

$\frac{ab+a+b+1}{1-a-b+ab}=\frac{a+b+1+ab}{1+ab-a-b}$

Both are always equal

But $\frac{x+1}{1-x}>0$

$\mathrm{x}\in(-1,1)\quad\therefore \mathrm{a}\in(-1,1),\mathrm{b}\in(-1,1)$

Q.7 Answer:

None of these

(where [.] and \{.\} denotes greatest integer function and fractional part function respectively)

Solution:

$f(x)=\log_{x-1}(-([x]+2))([x]-1))$

$([\mathrm{x}]+2)([\mathrm{x}]-1)<0;\mathrm{x}-1>0:\mathrm{x}-1\neq+1$

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$[x] \in (-2, 1), x > 1 : x \neq 2$

$x \in [-1, 1]$

$\therefore$ **Domain is** ϕ

$g(x) = \cos^{-1}(2 - \{x\})$

$-1 \leq 2 - \{x\} \leq 1$

$-3 \leq -\{x\} \leq -1$

$1 \leq \{x\} \leq 3 \Rightarrow x \in \phi$

$h(x) = \ln \ln(\cos x) = \ln \cos x > 0$

$\cos x > 1$

$\mathrm{x} \in \phi$

Q.8 Answer:

$\frac{2\pi}{3}$

Solution:

We obseve that the periods of $\sin 3x, |\sin 3x|$ **and** $[\sin 3x]$ **are** $\frac{2\pi}{3}, \frac{\pi}{3}$ **and** $\frac{2\pi}{3}$ **respectively.**

$\therefore$ **Period of** $f(x)$ **is** $\frac{2\pi}{3}$

Q.9 Answer:

Domain of $\mathrm{f(x)}$ is $[-1, 1]$

Solution:

$f(x) = g(x) + h(x)$

$f(x) = \frac{1}{\pi}\left(\sin^{-1} x + \tan^{-1} x\right) + \frac{1}{(x+1)+\frac{4}{x+1}}$

(A) Domain of $\mathrm{f(x)}$ **is** $[-1, 1]$

(B) Since $\sin^{-1} x$ **and** $\tan^{-1} x$ **are increasing functions,** $\max$ **will be at** $x = 1$

Maximum value of $g(x)$ **is** $g(1) = 3/4$

(C), Maximum value of $h(x)$ **is** $h(1) = 1/4$

Using AM-GM inequaltiy

(D) Range of $\mathrm{f(x)}$ **is** $[\mathrm{f}(-1), \mathrm{f}(1)]$

$\Rightarrow \left[-\frac{3}{4}, 1\right]$

Q.10 Answer:

Domain of $f(x)$ is R

Solution:

$f(x) = \log_2\left(\sin\left(x - \frac{\pi}{4}\right) + 3\right)$

so domain of $\mathrm{f(x)}$ **is** $\mathrm{x} \in \mathrm{R}$

Range $\rightarrow 2 \leq \sin\left(x - \frac{\pi}{4}\right) + 3 \leq 4$

so, $1 \leq \log_2\left(\sin\left(x - \frac{\pi}{4}\right) + 3\right) \leq 2;$

Range $\in [1, 2]$

Q.11 Answer:

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If $f(x)$ and $g(x)$ are one-one then $f(x)+g(x)$ is also one-one

Solution:

Let $h(x)=f(x)+g(x)$

$h'(x)=f'(x)+g'(x)$

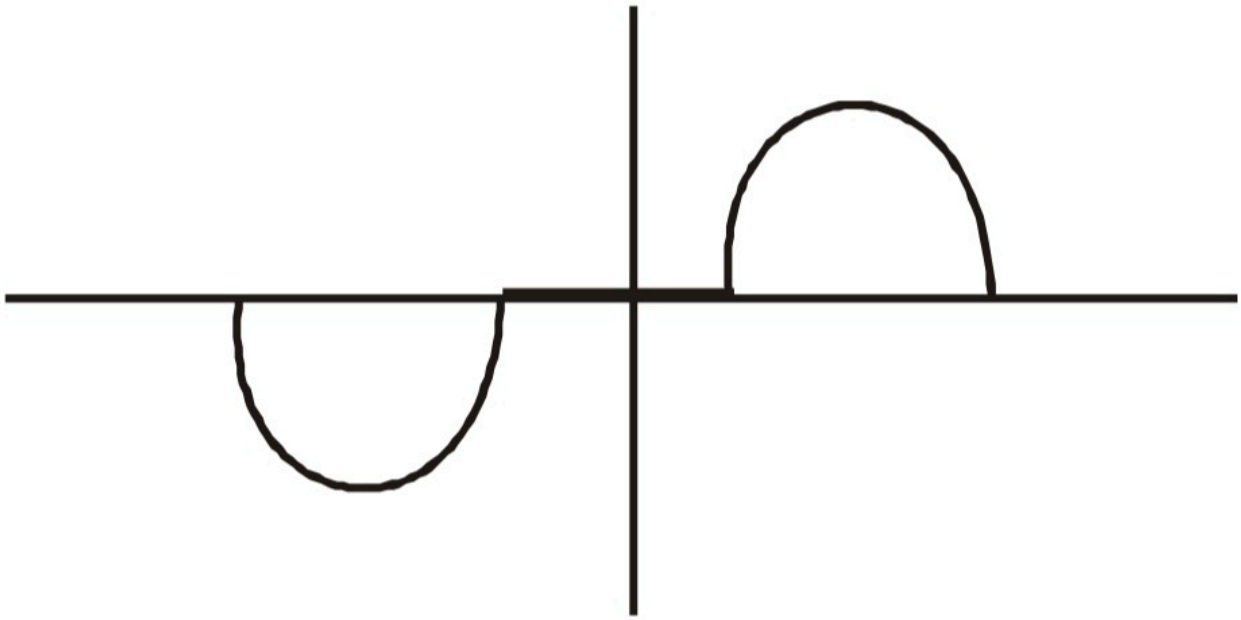
$h'(x)$ **may be +ve or** $-ve$

Let $\mathrm{I}(\mathrm{x})=\mathrm{f}(\mathrm{x})\cdot g(\mathrm{x})$

$I'(x)=f'(x)g(x)+f(x)g'(x)$

$\mathrm{I}'(\mathrm{x})=$ **may be** $+\mathrm{ve}$ **or** $-\mathrm{ve}$

This function is odd but not one-one



Q.12 Answer:

14

Solution:

$\phi(x)$ **is odd, one-one and onto**

So, $\mathrm{k}(\phi(\mathrm{x}))=-1+3+5+7=14$

Q.13 Answer:

18

Solution:

h **(x) is even, aperiodic, many-one and into**

So $\mathrm{k}(\mathrm{h}(\mathrm{x}))=0+4+6+8=18$

Q.14 Answer:

many-one into

Solution:

$\mathrm{k}\left(\mathrm{x}^2\right)=18, \quad \mathrm{k}\left(\mathrm{e}^{\mathrm{x}}\right)=19,$

$\mathrm{k}(\sin \mathrm{x})=16, \mathrm{k}(|\mathrm{x}|)=18$

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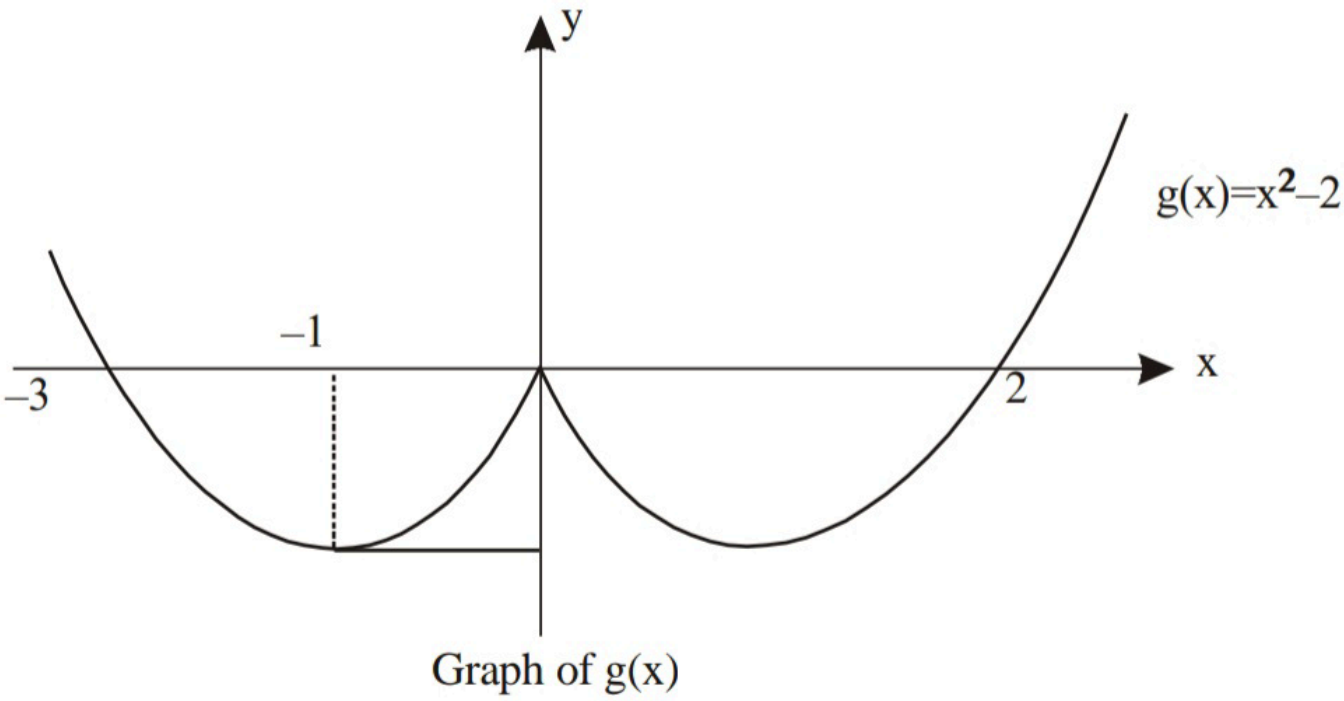
$\therefore$ $k(x)$ **is many-one into function**

Q.15 Answer:

$[-1, 3]$

Solution:

$f(x) = x^2 - 2|x|$



Graph of g(x)

Graph of $g(x)$

$g(x) = x^2 + 2x$

$$g(x) = \begin{cases} x^2 + 2x, -3 \leq x < -1 \\ -1, -1 \leq x < 0 \\ 0, 0 \leq x \leq 2 \\ x^2 - 2x, 2 < x \leq 3 \end{cases}$$

Range of g **(x) for** $[-3, -1]$ **is** $[-1, 3]$

Q.16 Answer:

3

Solution:

$f(x)$ **is not differentiable at** $x = 0$,
$\frac{dy}{dx} = 0$ **at** $x = \pm 1$

Q.17 Answer:

$x^2 + 2x + 1$

Solution:

For $x \in (-1, 0), f(x) - g(x) = x^2 + 2x + 1$

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Q.18 Answer:

A-p,r ; B-p,s,t ; C-q,s ; D-q,s

Solution:

(A) - pr, (B) - pst, (c) qs, (d) qs

(A) $f'(x) = \frac{\cos x}{2\sqrt{\sin x}}$ **if** $x \in \left[0, \frac{\pi}{3}\right]$ $f'(x) > 0$

$\Rightarrow$ **one-one also** $0 \leq f(x) \leq \left(\frac{\sqrt{3}}{2}\right)^{1/2}$ $\Rightarrow$ **not onto**

(B) $f'(x) = \frac{-4}{(x-1)^2} < 0$ **hence one-one also** $1 < f(x) < \infty \Rightarrow$ **onto**

(C) Clearly many-one, onto

(D) $f'(x) = \frac{x^2}{x-2}$

$\Rightarrow f'(x) = \frac{x^2-4x}{(x-2)^2}$

$\therefore \quad f'(x) < 0$ **if** $2 < x < 4$

$f'(x) > 0$ **if** $x > 4$

hence many one also $8 \leq f(x) \leq \infty$ **hence onto**

Q.19 Answer:

—

Solution:

Domain of $\sin^{-1}(\sin x)$ **is whole of** $R \ -\log_{\left(\frac{x+4}{2}\right)} \log_2 \left(\frac{2x-1}{3+x}\right) > 0$

i.e. $\log_{\left(\frac{x+4}{2}\right)} \left(\log_2 \left(\frac{2x-1}{3+x}\right)\right) < 0$

Case I : $0 < \frac{x+4}{2} < 1$

i.e. $-4 < x < -2$ **then** $\log_2 \frac{2x-1}{3+x} > 1$

i.e. $\frac{2x-1}{3+x} > 2$

i.e. $\frac{2x-1-6-2x}{3+x} > 0$

i.e. $x + 3 < 0$

i.e. $x < -3 \therefore -4 < x < -3.$

Case II : If $\frac{x+4}{2} > 1$

i.e. $x > -2$ **then** $0 < \log_2 \frac{2x-1}{3+x} < 1$

i.e. $1 < \frac{2x-1}{3+x} < 2$

i.e. $\frac{2x-1-3-x}{x+3} > 0$ **and** $\frac{2x-1-6-2x}{x+3} < 0$

i.e. $\frac{x-4}{x+3} > 0$ **and** $\frac{-7}{x+3} < 0$

i.e. $\{x < -3$ **or** $x > 4\}$ **and** $x > -3$

i.e. $x > 4$

From eq. (1) and (2), $x \in (-4, -3) \cup (4, \infty)$

$\therefore \quad a = -4, b = -3, c = 4$ **and** $s0, a + b + 3c = 5$

Q.20 Answer:

—

Solution:

$f : (0, \infty) \to (0, \infty)$

$f(xf(y)) = x^2 y^a (a \in R)$

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Put $x = 1$, **we get** $f\left(f(y)\right) = y^a$

Put $f(y) = \frac{1}{x}$, **we get** $f(1) = \frac{1}{(f(y))^2} \cdot y^a$ **put** $y = 1$ **we get** $(f(1))^3 = 1$

$\therefore$ $f(1) = 1$ **for** $y = 1$, **we have** $f(x(f(1))) = x^2$

$\therefore$ $f(x) = x^2$

$f(xf(x)) = x^2x^a$

$f\left(x^3\right) = x^2x^a$

$f\left(x^3\right) = x^6$

$\therefore$ $x^6 = x^2x^a$

thus $a = 4$

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